Incomplete Multi-View Drug Recommendation via Multi-Level Representation Learning and Curriculum Learning

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Abstract

The drug recommendation task aims to provide effective and safe prescription decision support for clinical treatment based on patients' past Electronic Health Records (EHR). However, the prevalent phenomenon of missing views in multi-source heterogeneous EHR data may cause performance degradation. This is due to the lack of sufficient information and increased learning difficulties, which limit the practical effectiveness of drug recommendation models in medical applications. In this paper, we emphasize the problems of incompleteness in practical drug recommendation and propose the **I**ncomplete **M**ulti-View **D**rug **R**ecommendation model via Multi-Level Representation Learning and Curriculum Learning named **IMDR**. In particular, IMDR employs a Multi-Level Representation Learning architecture equipped with a Medical Code-Level Drug Knowledge Infusion Module and a Visit-Level Cross-View Information Module for patient representation learning to overcome the information loss caused by incomplete data. And then, a Gaussian-guided curriculum learning strategy is proposed to assist the learning process of IMDR with a novel difficulty measure to achieve effective progressive learning under missing medical views. Systematic evaluation on two large-scale real-world medical datasets, MIMIC-III and MIMIC-IV, demonstrates that IMDR reduces the Drug-Drug Interaction (DDI) rate by 2.97% compared to existing state-of-the-art drug recommendation baselines, while achieving significant improvements of 3.29% and 1.97% in Jaccard similarity scores and F1 score, respectively. Furthermore, compared to advanced incomplete multi-view learning (IML) models, IMDR's advantages in Jaccard similarity scores and F1 score further expand to 4.03% and 2.41%.

CCS Concepts

• **Applied computing** → **Health informatics**; • **Information systems** → *Data mining*.

Keywords

Drug recommendation, Healthcare, Incomplete multi-view learning

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1 Introduction

In recent years, intelligent drug recommendation systems have made significant progress in supporting clinical decision-making [2]. As shown in Figure 1(a), these systems primarily integrate multi-view medical entity representations (including disease diagnoses, surgical procedures, and clinical symptoms) to recommend personalized drug combination schemes for patients while avoiding potential DDI risks [22, 28].

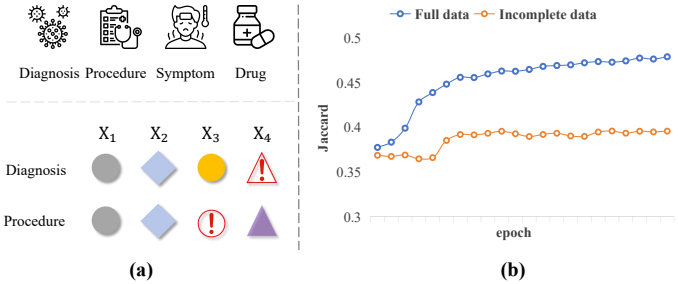


Figure 1: (a) The figure shows typical EHR views (top) and sample alignment examples (bottom), where X_1 and X_2 are complete samples while X_3 and X_4 contain missing views. (b) Performance comparison of GAMENet shows significantly lower Jaccard scores when trained on incomplete multi-view data compared to full data, with the performance gap persisting throughout the training process.

However, in actual clinical environments, data from different domains, sensing devices, or feature extractors often exhibit partial missing patterns, mainly attributed to technical factors such as data storage system failures and device instability [5, 27]. Taking EHR as an example, its multi-view data encompassing disease diagnoses, surgical procedures, and clinical symptoms serves as crucial input for drug recommendation systems. However, Imbalanced distribution of medical resources, dynamic adjustments in medical protocols and restrictions on inter-institutional data access often result in the loss of critical clinical information[29]. Besides, data privacy concerns and other technical failures hinders the completeness of clinical data[7]. The interaction of these factors leads to prevalent incompleteness in multi-view data used for drug recommendations [15], severely limiting the practical application effectiveness of deep learning technologies in healthcare and making it difficult to achieve the expected revolutionary goals. Therefore, ensuring the prediction accuracy and clinical safety of drug recommendation models under conditions of missing medical view data has become a critical scientific problem requiring urgent resolution.

Our work focuses on maintaining recommendation accuracy and safety in drug recommendation models under scenarios with missing medical views. To achieve this objective, our work faces two major challenges: **1) How to overcome the lack of sufficient information for patient representation learning arising from incomplete view medical data.** Multiple views of medical codes provide complementary information, but existing models are predominantly designed based on complete data assumptions. Under incomplete view inputs, this results in a reduction of information during patient representation learning. As shown in the Figure 1(b), recommendation performance is severely degraded when dealing with incomplete views. **2) How to alleviate the increased learning difficulty under heterogeneous data conditions with missing views.** Owing to the intricate patterns exhibited by patients, existing models encounter substantial challenges in extracting drug-related information even through diverse network designs. The incompleteness of medical codes exacerbates the learning difficulty, thereby reducing the effectiveness of these drug recommendation models. Therefore, designing efficient training algorithms adapted to incomplete view scenarios becomes crucial in real medical situations.

To address these challenges, this paper proposes IMDR[1], a drug recommendation model suitable for incomplete multi-view scenarios. For drug recommendation, the accuracy of drug-related information in patients' representations can affect the final performance. For cases with missing views, the learned patient representations may not provide sufficient information. To overcome this issue, IMDR employs an innovative fine-grained multi-level representation learning architecture for accurate patient representation learning. Specifically, we design a code-level drug knowledge fusion module to inject drug-related information into medical codes and a Visit-level Cross-view Information Alignment module for aligning the knowledge among different views for learning accurate patient representation. To minimize the learning difficulty, IMDR is designed with a novel curriculum learning paradigm to guide the model learning from easy to hard. While traditional models typically rely on complete view data for initial training to ensure convergence, we aim to simultaneously process both complete and incomplete view samples, achieving joint optimization and effectively characterizing heterogeneous sample distributions. Therefore, we develop an adaptive curriculum learning mechanism based on Gaussian distribution that automatically adjusts sample difficulty weights to enable progressive learning from simple to complex samples. In summary, the main contributions of this work are as follows:

- We are the first to introduce the IML paradigm to the field of drug recommendation, proposing a solution that more closely aligns with real clinical environments. We propose a Incomplete Multi-View Drug Recommendation model via Multi-Level Representation Learning and Curriculum Learning, namely IMDR to handle incompleteness of medical codes.
- To overcome the lack of sufficient information, IMDR learns accurate patient representations with fine-grained multi-level representation learning via both code-level drug knowledge fusion and visit-level cross-view information alignment to learn efficient patient representation under incomplete views.
- To reduce learning difficulty under missing views, we introduce a Gaussian-guided curriculum learning strategy with a novel complex difficulty measurement to assist robust progressive training, guiding models to learn patient representations from the easy to hard way.
- Our extensive experiments on two large-scale real-world EHR datasets, MIMIC-III and MIMIC-IV, demonstrate that our IMDR exhibits excellent recommendation accuracy and medication safety in incomplete view scenarios. Besides, in our comparative experiments with other IML models, IMDR demonstrates superior adaptability in drug recommendation task.

2 RELATED WORK

2.1 Drug Recommendation

Existing drug combination recommendation methods can be categorized into instance-based and longitudinal-based drug recommendation. Instance-based methods focus on patients' current health

[1]Code is available at: https://github.com/Tyunsen/imdr

status. For example, LEAP [28] extracts feature information from current visit records and employs a multi-instance multi-label drug recommendation setting. However, most studies [22] advocate for longitudinal approaches to leverage temporal dependencies in clinical histories. Among these, RETAIN [3] utilizes a dual-layer RNN with attention mechanisms to model longitudinal information and identify the most significant historical diagnoses, thereby making the model interpretable. GAMENet [16] further advances sequence modeling based on GRU, employing memory neural networks and storing historical medication information as reference for subsequent predictions. SafeDrug [25] incorporates drug molecular structure information, utilizing a graph-based molecular encoder with message passing neural network (MPNN) and learnable fingerprints. Additionally, MICRON [24] deviates from using GRU encoders, instead focusing on learning medication change patterns between different visits, avoiding RNN for longitudinal sequence modeling to achieve more efficient patient longitudinal representation. Recently, MoleRec [26] proposed an attention learning method that models molecular substructure interactions and correlations between patient conditions and substructures, improving both accuracy and safety of drug combination recommendations. The latest innovation, REFINE [1], further advances this approach by not only considering the severity of drug interactions but also explicitly modeling drug dosage adjustments and laboratory test result trends. This fine-grained modeling approach enables REFINE to provide more personalized and safer drug combination recommendations.

2.2 Incomplete Multi-View Learning

With the widespread application of multi-view data, incomplete multi-view learning has emerged as a significant research direction in machine learning. Early work primarily employed statistical methods, such as zero-filling and mean-filling strategies, to handle missing views [6]. While these methods are simple to implement, they often neglect the intrinsic correlations between views, making it difficult to ensure the quality of completed data. With the advancement of deep learning, IML methods have been categorized into discriminative and generative approaches. Discriminative methods primarily encompass matrix learning and graph learning. In matrix learning, PVC[9] proposed using non-negative matrix factorization for consistent representations, lrMMC[13] introduced low-rank matrix decomposition, and JELLA[17] combined embedding learning for representation alignment. In graph learning, PIC[20] utilized Laplacian graphs for common representations, while IMSC_AGL[21] integrated spectral clustering with adaptive graph learning. For generative approaches, DCP[10] proposed dual contrastive prediction mechanisms, CRA[18] developed cascaded residual autoencoders, and COMPLETER[11] exploited inter-view consistency through contrastive learning. Recent advances include IMVCDC[4] employing diffusion models and VIST[14] implementing a dual-layer Transformer with KNN and Gaussian distribution-based generation. Current research trends focus on integrating multiple methodological advantages, as exemplified by LMVCAT [12] combining probabilistic losses with deep models. Future research priorities include enhancing model robustness, reducing computational complexity, and exploring new learning paradigms.

3 Preliminaries

This section formulates the problem of learning a drug recommendation model that generates safe drug combinations by considering a patient's incomplete multi-view EHR data.

3.1 Definitions

3.1.1 Incomplete multi-view EHR data. Given EHR data $\{(X_n, Y_n)\}_{n=1}^{N}$ for N patients, each patient's complete medical history is represented as multi-view temporal data, including medical code sequences such as diagnosis, procedure, and medication records. For simplicity of exposition, we describe our method using a single patient as an example, omitting the patient index n. Formally, a patient's first T EHR visits can be represented as a sequence $X_T = \{X_T^{(v_i)}\}_{i=1}^{2}$, where $X_T^{(v_1)}$ represents the patient's diagnoses view sequence data for the first T visits, and $X_T^{(v_2)}$ represents the patient's procedure view sequence data for the first T visits. For each sequence data $X_T^{(v_i)} = [c_0^{(v_i)}, c_1^{(v_i)}, ..., c_t^{(v_i)}, ..., c_T^{(v_i)}]$, $c_t^{(v_i)} \in \{0,1\}^{|\mathcal{D}^{(v_i)}|}$ is the multi-hot encoding vector of diagnoses or procedures during the patient's t-th hospitalization, where $\mathcal{D}^{(v_i)}$ represents the corresponding set of diagnoses and procedures. Correspondingly, the drug set labels for the patient's first T visits are denoted as $Y_T = [l_0, l_1, \ldots, l_t, \ldots, l_T]$, where $l_t \in \{0,1\}^{|\mathcal{D}^{\mathcal{L}}|}$ is the multi-hot encoding vector of the drug set for the t-th hospitalization, and $\mathcal{D}^{\mathcal{L}}$ represents the drug set. In the missing view scenario, we introduce an index matrix $M \in \mathbb{R}^{N\times 2}$ to represent view availability.

$$M_n^{(v_i)} = \begin{cases} 1, & \text{the } v_i \text{ view of the } n\text{-th sample is observable} \\ 0, & \text{otherwise} \end{cases}$$

Each sample contains at least one view, and the degree of view missingness in the dataset is represented by the missing rate $\eta = \frac{\widetilde{N}}{N}$, where $\widetilde{N}$ represents the number of samples with missing views.

3.1.2 Safe Drug Recommendation. This work not only addresses the drug recommendation problem in multi-view missing scenarios but also aims to control the risk of DDI in recommended drug combinations. We use a symmetric binary adjacency matrix $A^{(d)} \in \mathbb{R}^{|\mathcal{D}^{\mathcal{L}}|\times|\mathcal{D}^{\mathcal{L}}|}$ to represent DDI relationships. When $A^{(d)}[j,k] = 1$, it indicates a potential interaction risk between drugs j and k, while $A^{(d)}[j,k] = 0$ indicates that the two drugs can be safely used in combination.

3.2 Problem Formulation

This work aims to learn a drug recommendation model $f(\cdot)$ that, under a missing rate of η, can generate drug combination recommendations for any patient with complete or incomplete medical records at their t-th visit, given the patient's visit sequence $X_t^{(v_i)} = \{M_n^{(v_i)} * X_t^{(v_i)}\}_{i=1}^{2}$ up to the current visit, and the target drug combination l_t, as well as the DDI matrix $A^{(d)}$, $f(\cdot)$ can generate a drug recommendation combination:

$$\hat{l}_t = f\left(\{M_n^{(v_i)} * X_t^{(v_i)}\}_{i=1}^{2}\right)$$

During training, the model's objective function consists of at least two components: (i) supervised learning loss based on actual medication records l_t for $\hat{l}_t \in \mathbb{R}^{|\mathcal{D}^{\mathcal{L}}|}$; (ii) drug safety constraint loss

based on the DDI adjacency matrix $A^{(d)}$, used to control DDI risks in the recommendation results.

4 METHOD

This section outlines the methodology of IMDR. As illustrated in Figure 2, IMDR features a novel architecture combining two key levels: drug knowledge integration at the medical code-level and information alignment at the visit level. This structure is further enhanced by an adaptive curriculum learning strategy guided by Gaussian distribution, enabling progressive model training.

4.1 Medical Code-level Drug Knowledge Infusion Module

In this section, we will introduce the details the first part of the multi-level representation learning architecture called the Medical Code-level Drug Knowledge Infusion Module, which includes the description of Embedding Layer, Drug Representation Network and Code-level Drug Knowledge Fusion.

4.1.1 Embedding Layer. To fully capture the rich feature information in multi-view temporal data, we construct an embedding table $\mathcal{E} \in \mathbb{R}^{|\mathcal{D}|^{(v_i)} \times d}$ with dimension d for view i. In scenarios with missing views, effective information fusion is crucial for enhancing model robustness. To integrate drug prior knowledge into ICD codes, we first extract the valid token matrix for each visit:

$$\mathcal{I} = \{j \mid c_t^{(v_i)}[j] = 1\}$$

$$e_{\mathcal{I}}^{(v_i)} = \left[\mathcal{E}[\mathcal{I}_1], \mathcal{E}[\mathcal{I}_2], ..., \mathcal{E}[\mathcal{I}_k]\right]^T, k = |\mathcal{I}|$$

where $c_t^{(v_i)} \in \{0,1\}^{|D^{(v_i)}|}$ represents the multi-hot encoding of view v_i during the patient's t-th visit. $e_{\mathcal{I}}^{(v_i)} \in \mathbb{R}^{|\mathcal{I}| \times d}$ denotes the embedding feature matrix for this visit, with each row $e_{\mathcal{I}}^{(v_i)}[\mathcal{I}_j]$ corresponding to the embedding vector of the $\mathcal{I}_j$-th code.

4.1.2 Drug Representation Network. IMDR employs graph neural networks (GNN) to encode drugs, updating and obtaining drug representations through iterative aggregation of neighboring node information. The process is as follows:

$$a^{(k)}(u) = MESSAGE^{(k)}\left(z_n^{(k-1)}(v), z_e(u,v) \middle| v \in N_{A^{(*)}}(u)\right)$$

$$z_n^{(k)}(u) = UPDATE^{(k)}(a^{(k)}(u), z_n^{(k-1)}(u))$$

where $z_n^{(k)}(u) \in \mathbb{R}^d$ represents the node representation of drug u at the k-th layer, $z_e(u,v)$ denotes the edge features between drugs u and v, and $N_{A^{(*)}}$ represents the neighbor set of drug u in the adjacency matrix $A^{(*)}$.

Here, $A^{(*)}$ includes both the DDI adjacency matrix $A^{(d)}$ and co-occurrence matrix $A^{(c)}$, capturing drug-drug interactions and co-prescription patterns respectively. By stacking $z_n^{(k)}(u)$, we obtain two types of drug representations: $Z_{A^{(d)}} \in \mathbb{R}^{|\mathcal{D}^{\mathcal{L}}| \times d}$ from the DDI graph and $Z_{A^{(c)}} \in \mathbb{R}^{|\mathcal{D}^{\mathcal{L}}| \times d}$ from the co-occurrence graph, where both are learned using graph convolutional networks (GCNs). The final drug representation is obtained through a weighted combination:

$$Z = Z_{A^{(d)}} + \omega Z_{A^{(c)}}$$

where ω is a learnable parameter that balances the contribution of interaction-based and co-occurrence-based features.

4.1.3 Code-level Drug Knowledge Fusion. Our work proposes a code-level knowledge fusion mechanism specifically designed for scenarios with missing views. Traditional methods typically integrate drug features after generating patient representations to obtain drug representations that better align with patients' health conditions. However, in scenarios with missing views, delayed knowledge injection may fail to effectively compensate for information loss. Therefore, we propose incorporating fine-grained code-level drug knowledge into each ICD code during the initial stage of code information transmission, thereby establishing model robustness early on and effectively addressing the missing view problem.

$$e_t^{(v_i)} = \sum_{j=0}^{|\mathcal{I}|} \left(e_{\mathcal{I}}^{(v_i)}[\mathcal{I}_j] + SOFTMAX\left(\frac{e_{\mathcal{I}}^{(v_i)}[\mathcal{I}_j] Z^T}{\sqrt{d}}\right) Z \right)$$

Finally, we perform element-wise summation of all embedding vectors within the visit to obtain the view features $e_t^{(v_i)} \in \mathbb{R}^d$, which representing the patient's t-th visit.

This code-level knowledge fusion strategy enables embedding vectors to effectively integrate external knowledge during the training process. Even when facing missing views during the inference phase, the model can still make accurate predictions based on the previously acquired knowledge.

4.2 Visit-level Cross-view Information Alignment

This module aims to achieve information alignment and interactive fusion between multiple views at the visit level. This module adopts a three-stage architecture: 1) first, it captures temporal dependencies of each view through two shared-weight self-attention encoder, 2) then facilitates information interaction between views using a cross-attention encoder, 3) and finally aligns the representation distributions of different views through contrastive learning methods.

4.2.1 Bi-View Sequential Representation Learning. In the view representation module, we must not only understand patients' temporal health progression but also address the information loss in missing view scenarios. Therefore, it is crucial to capture both the temporal information across different visits of the same patient and the multi-view information at the same visit. Specifically, we first employ two shared-weight self-attention encoders to establish temporal relationships for each view, followed by a cross-attention encoder to capture consistency information between dual views.

To capture view information at different time steps, following [19], we augment the view embedding vector for the t-th visit with positional encoding:

$$pe_{(t,2k)} = sin(t/10000^{2k/d})$$

$$pe_{(t,2k+1)} = cos(t/10000^{2k/d})$$

$$E_t^{(v_i)} = \left[e_1^{(v_i)} + pe(1), e_2^{(v_i)} + pe(2), ..., e_t^{(v_i)} + pe(t)\right]^T$$

Here k is an index enumerating positions in the embedding dimension. Through the introduction of positional encoding, we can

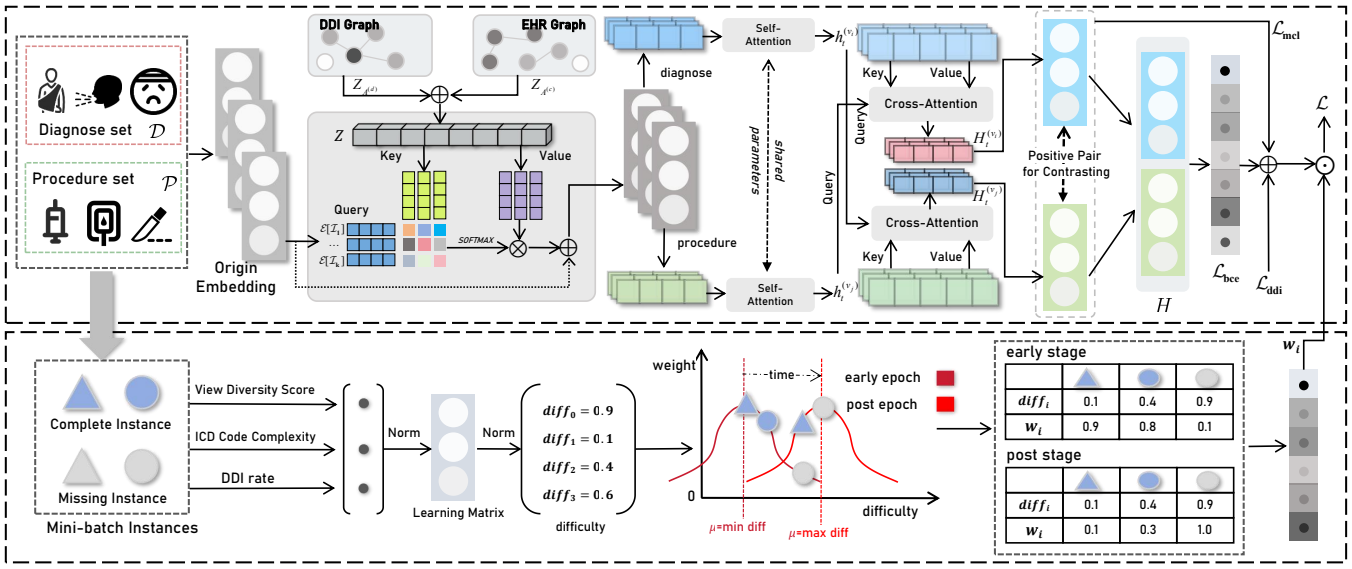


Figure 2: Overall architecture of the IMDR model. During dual-level representation learning, IMDR enriches medical code embeddings $e_t^{(v_i)}$ with drug knowledge Z at the code-level, then aligns visit-level representations $H_t^{(v_i)}$ and $H_t^{(v_j)}$ through cross-view attention and contrastive learning to handle missing views. Meanwhile, a Gaussian-guided curriculum learning strategy with dynamic distribution parameters μ and σ guides the training process from simple samples (complete views, low DDI rates) to complex samples (missing views, high clinical complexity).

incorporate temporal visit information into the view embeddings. Furthermore, we can represent a patient's view information at different time steps as a matrix $E_t^{(v_i)} \in \mathbb{R}^{t \times d}$. Subsequently, we utilize a Transformer Encoder to process this temporal sequence and obtain hidden view representations:

$$h_t^{(v_i)} = Transformer(Q = E_t^{(v_i)}, K = E_t^{(v_i)}, V = E_t^{(v_i)})$$

After obtaining the hidden representations of each view through the inter-view self-attention mechanism, we aim to further integrate information between different views:

$$H_t^{(v_i)} = Transformer(Q = h_t^{(v_i)}, K = h_t^{(v_j)}, V = h_t^{(v_j)})$$

In the above equation, v_j represents a view different from (v_i). This sub-layer employs a cross-attention mechanism to ensure thorough information fusion between different views, thereby better representing the overall condition of the same patient. In summary, we can obtain a temporal view representation matrix $H_t^{(v_i)} \in \mathbb{R}^{t \times d}$ for a patient.

4.2.2 Visit-level Multi-View Contrastive Learning. In patient EHR data, both diagnosis and procedure information represent the same patient's health status, containing rich complementary information and consistency between them. In scenarios with missing views, encoding patients using only single-view data can easily lead to inconsistency in the final patient representations. Therefore, we employ contrastive learning to align the distribution of patient diagnosis and procedure in the representation space, thereby obtaining more consistent patient representations.

$$\mathcal{L}_{mcl} = -log \frac{exp(sim(x, x^+)/\tau)}{exp\,(sim(x, x^+)/\tau) + \sum_{x^- \in Neg(x)} exp(sim(x, x^-)/\tau)}$$

$$sim\,(x, x^+) = \frac{xx^+}{||x||||x^+||}$$

Here, $x \in \mathbb{R}^d$ denote a representation at a certain visit from $H_t^{(v_i)}$, $x^+ \in \mathbb{R}^d$ represent another view's representation at the same visit from $H_t^{(v_j)}$, x^- represents representations from different time step within the same mini-batch, where $\mathcal{N}eg(x)$ is the set of negative samples, $sim(\cdot, \cdot)$ denotes the cosine similarity, and τ is the temperature parameter for scaling the contrastive loss.

4.2.3 Drug Prediction Layer. To generate a more compact patient representation, we concatenate the view representations H_t^v into a patient representation matrix. We then apply a feed-forward neural network $fc_1(\cdot) : \mathbb{R}^{2d} \rightarrow \mathbb{R}^d$ to obtain the patient representation H:

$$H = fc_1([H_t^{(v_i)}[CLS] : H_t^{(v_j)}[CLS]])$$

In the above equation, we only extract the most recent visit $H_t^{(v_i)}[CLS]$ to represent the patient's recent health status view representation. Subsequently, we perform multi-label drug prediction through another feed-forward neural network $fc_2(\cdot) : \mathbb{R}^d \rightarrow \mathbb{R}^{|\mathcal{D}^L|}$ with $|\mathcal{D}^L|$ output nodes:

$$\hat{l}_t = \sigma(fc_2(H))$$

where σ is the sigmoid activation function, and the k-th element of $\hat{l}_t \in \mathbb{R}^{|\mathcal{D}^L|}$ represents the probability of recommending the k-th drug during the t-th visit. We treat the prediction of each drug as a subtask of drug combination recommendation and employ the commonly used Binary Cross-Entropy (BCE) loss function. The

supervised loss for this sample can be expressed as:

$$\mathcal{L}_{bce} = -\sum_{k=1}^{|\mathcal{D}^L|} [l_{t,k} \log(\hat{l}_{t,k}) + (1 - l_{t,k}) \log(1 - \hat{l}_{t,k})]$$

where l_t denote the target labels.

4.3 Gaussian-guided Curriculum Learning

In this section, to reduce the learning difficulty and achieve progressive learning, we develop an adaptive curriculum learning mechanism based on Gaussian distribution that automatically adjusts sample difficulty weights. In fact, there are two main design perspectives we should consider during the design of curriculum learning: **Q1**. How to construct and quantify difficulty metrics for patient training samples? **Q2**. How to formulate appropriate dynamic learning strategies to optimize network parameters?

To systematically evaluate sample difficulty, we propose a three-dimensional difficulty assessment metric. Given a patient's multi-hot encoding for the t-th visit $c_t^{(v_i)}$ and corresponding ground truth l_t, we measure the sample's difficulty level from three dimensions: view completeness score, drug interaction complexity, and clinical semantic complexity (Q1).

$$s_v = M_n^{(v_i)}(1 + \epsilon)$$

Here s_v represents the view completeness score, where $M_n^{(v_i)} \in \mathbb{R}^{N\times 2}$ denotes the view availability matrix, and ϵ is a balancing factor. s_v reflects the completeness of view information in the sample, with higher scores indicating more available view information.

$$s_d = \frac{l_t^T(A^{(d)} \odot (l_t l_t^T))l_t - diag(l_t^T A^{(d)} l_t)}{l_t^T l_t (l_t^T l_t - 1)/2}$$

Here s_d represents the drug interaction complexity derived from DDI rate, which quantifies the complexity of interactions within drug combinations. Higher DDI rates indicate more severe drug conflicts in the sample. We designed this metric to enable the network to distinguish samples with different DDI levels during different training periods.

$$s_c = Norm(log(1 + \rho|c_t^{(v_i)}|)\mathcal{F}_{lm}(c_t^{(v_i)}))$$

Finally, s_c represents the clinical semantic complexity. Here, $\mathcal{F}_{lm}$ refers to any reliable language model, and we used GPT-4o to design prompt templates for three medical entity encodings. More information about prompts can be found in Appendix A.

Through these prompts, s_c quantifies medical entities across multiple dimensions including "medication complexity", "side effect profile", "disease severity", "complication risk", "technical difficulty", and "required resources". The final score is weighted based on the current entity set length $|c_t^{(v_i)}|$, where ρ is a length balancing factor, and $Norm(.)$ represents min-max normalization within a mini-batch. Based on these metrics, the overall difficulty of a clinical sample is defined as:

$$diff = Norm(SOFTMAX(W^T)\,[s_v : s_d : s_c]^T)$$

where $W \in R^3$ is a learnable weight matrix. Next, to achieve smooth curriculum learning (Q2), we design a binary Gaussian function to dynamically allocate sample loss weights. First, we define the dynamic center point μ and standard deviation σ:

$$\mu = diff_{\min} + (diff_{\max} - diff_{\min}) \cdot t, \sigma = 0.5(1 + \delta - t)$$

where $t \in [0, 1]$ represents the normalized training progress (the ratio of current epoch to total training epochs). $diff_{min}$ and $diff_{max}$ represent the lower and upper bounds of sample difficulty in the current mini-batch, respectively, and δ is a positive hyper-parameter. As the training progresses, the dynamic center point gradually transitions from $diff_{min}$ to $diff_{max}$, while the standard deviation σ gradually contracts, making sample selection more focused. Based on these parameters, the sample weight is calculated through the following function:

$$\omega(diff, t) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{(diff - \mu)^2}{2\sigma^2}\right)$$

During training, samples closer to the current center point receive larger weights, thereby implementing a dynamic curriculum learning strategy. Specifically, the network focuses on simple samples during the initial training phase, gradually shifts attention to more challenging samples as training progresses, and ultimately focuses on the most complex samples in the final training phase. We incorporate this dynamic weighting strategy into the supervised learning objective by modifying the BCE loss as follows:

$$\mathcal{L}_{bce} = -\omega(diff, t) \sum_{k=1}^{|\mathcal{D}^L|} [l_{t,k} \log(\hat{l}_{t,k}) + (1 - l_{t,k}) \log(1 - \hat{l}_{t,k})]$$

By evaluating sample difficulty through multi-feature combination and implementing smooth weight allocation through Gaussian distribution, this method can enhance the model's learning effectiveness, particularly when handling heterogeneous samples. Some more visual examples of our curriculum learning are provided in Appendix 5.5.

4.4 Training Objectives

In addition to the supervised learning loss function and multi-view contrastive learning loss, our model incorporates a DDI control loss to constrain drug combination recommendations. The DDI control loss serves as a constraint term specifically designed to reduce potential adverse interaction risks in recommended drug combinations. For drug combinations predicted by the model, we minimize their DDI risk through the following loss function:

$$\mathcal{L}_{ddi} = \sum_{j=1}^{|\mathcal{D}^L|} \sum_{k=1}^{|\mathcal{D}^L|} \left(A \odot \left(\hat{l}_t \hat{l}_t^T\right)\right)[j, k]$$

where $\hat{l}_t \hat{l}_t^T \in R^{|\mathcal{D}^L|*|\mathcal{D}^L|}$ represents the co-occurrence probability matrix of any two drugs in the same recommended combination. $\odot$ denotes the Hadamard product. If $A[j, k] = 1$, then there exists a potential interaction risk between drugs j and k.

During the training process, improvements in recommendation accuracy are often accompanied by increased DDI risks. This phenomenon primarily stems from the inherent drug interaction patterns present in real prescriptions within EHR data. Consequently, regardless of the model's prediction accuracy, it may lead to high

Table 1: Performance MIMIC-III and MIMIC-IV in terms of DDI rate, Jaccard, F1-score. The best and runner-up results are highlighted in bold and underline respectively.

Missing rate η		0.1			0.5			0.9			Avg		
Dataset	**Model**	**Jaccard ↑**	**F1-score ↑**	**DDI ↓**	**Jaccard ↑**	**F1-score ↑**	**DDI ↓**	**Jaccard ↑**	**F1-score ↑**	**DDI ↓**	**Jaccard ↑**	**F1-score ↑**	**DDI ↓**
MIMIC-III	Transformer	0.4986	0.6470	0.0590	0.4601	0.6125	0.0598	0.4154	0.5659	0.0693	0.4536	0.6037	0.0618
	MLP	0.4834	0.6327	0.0588	0.4566	0.6070	0.0613	0.4456	0.5960	0.0604	0.4606	0.6107	0.0607
	RETAIN	0.5010	0.6407	0.0560	0.4529	0.5952	0.0654	0.4398	0.5808	0.0694	0.4602	0.6014	0.0615
	GAMENet	0.4908	0.6392	0.0665	0.4521	0.6035	0.0584	0.4296	0.5826	0.0667	0.4537	0.6053	0.0640
	SafeDrug	0.4779	0.6230	**0.0553**	0.3950	0.5341	**0.0536**	0.3876	0.5407	0.0706	0.4062	0.5536	0.0642
	MICRON	0.4945	0.6439	0.0603	0.4677	0.6178	0.0613	0.4499	0.6005	0.0636	0.4678	0.6181	0.0619
	MoleRec	0.4629	0.6148	0.0572	0.4543	0.6075	0.0670	0.4384	0.5922	0.0591	0.4500	0.6030	0.0622
	IMDR	**0.5030**	**0.6497**	0.0638	**0.4809**	**0.6270**	0.0566	**0.4728**	**0.6218**	**0.0580**	**0.4832**	**0.6303**	**0.0597**
MIMIC-IV	Transformer	0.4979	0.6395	0.0570	0.4440	0.5857	0.0651	0.3855	0.5270	0.0619	0.4367	0.5773	0.0643
	MLP	0.4603	0.5994	0.0565	0.4353	0.5743	0.0575	0.4072	0.5464	0.0616	0.4345	0.5738	0.0587
	RETAIN	0.4974	0.6375	0.0559	0.4485	0.5908	0.0623	0.4396	0.5806	0.0606	0.4550	0.5966	0.0595
	GAMENet	0.4938	0.6331	0.0632	0.4547	0.6006	0.0631	0.4280	0.5746	0.0750	0.4585	0.6026	0.0661
	SafeDrug	0.4720	0.6080	0.0506	0.3806	0.5038	0.0555	0.2975	0.4043	0.0555	0.3829	0.5072	0.0573
	MICRON	**0.5020**	**0.6423**	0.0548	0.4657	0.6074	0.0584	0.3997	0.5443	0.0571	0.4614	0.6037	0.0576
	MoleRec	0.4084	0.5592	**0.0487**	0.3891	0.5419	0.0656	0.3801	0.5326	0.0608	0.3895	0.5423	0.0603
	IMDR	0.4921	0.6293	0.0565	**0.4754**	**0.6133**	**0.0534**	**0.4582**	**0.5966**	**0.0554**	**0.4750**	**0.6133**	**0.0556**

↓ means the corresponding metric is the lower the better and ↑ means the opposite.

DDI risks, presenting a significant challenge in balancing recommendation accuracy and medication safety. Inspired by [16], we introduce a DDI safety threshold as a constraint to ensure medication safety while maintaining recommendation accuracy. Accordingly, we construct the following comprehensive loss function:

$$\mathcal{L} = \begin{cases} \mathcal{L}_{bce} + \alpha \mathcal{L}_{ddi} + \beta \mathcal{L}_{mcl} & if\ ddi \geq \theta \\ \mathcal{L}_{bce} + \beta \mathcal{L}_{mcl} & if\ ddi < \theta \end{cases}$$

where ddi is the DDI score calculated based on the current prediction results $\hat{l}_t$, θ is our predetermined DDI threshold, and α and β are the parameters for the DDI control loss and multi-view contrastive learning loss, respectively.

5 Experiments

5.1 Dataset and Baselines

For dataset, we conduct experiments based on two public clinical dataset MIMIC-III and MIMIC-IV and performed unified data processing using the open-source benchmark framework PyHealth[2][23]. More dataset information can be found in Appendix B.1.

For baselines, we include two categories of previous works. One category falls into methods for drug recommendation like Multilayer Perception(MLP), **Transformer** [16],**RETAIN** [3],**GAMENet** [16],**SafeDrug** [25], **MICRON** [24], **MoleRec** [26]. All of these models are implemented in PyHealth[23]. The other categories fall into the recent incomplete multi-view learning methods including **zero-filling**, **mean-filling** [8], **View-copying**, **random-filling**, **DCP**[10], **IMVCDC** [4], and **VIST** [14]. More information about baselines can be found in Appendix B.2.

5.2 Performance Comparison with Advanced Drug Recommendation Baselines

To ensure the reliability and statistical significance of experimental results, we conducted comparative experiments using five different random seeds and reported the average performance across evaluation metrics. The analysis focuses on three representative scenarios with missing rates of 0.1, 0.5, and 0.9. To comprehensively evaluate the model's overall performance, we calculated the mean values of all metrics for missing rates ranging from 0.1 to 0.9, presented in the rightmost column of the table 1. It should be noted that since traditional drug recommendation models cannot directly handle missing view scenarios in their architectural design, we adopted zero-filling as the baseline strategy in our comparative experiments.

From an overall perspective from table 1, the IMDR model demonstrated exceptional comprehensive performance on both MIMIC-III and MIMIC-IV datasets. On average, the model outperforms the baseline method significantly in all evaluation metrics, particularly in terms of Jaccard similarity and F1-score. Notably, the drug combinations recommended by IMDR exhibited outstanding safety performance, with DDI rates significantly overall lower than comparison models and maintaining stability across different missing rates.

Further analysis reveals IMDR's robust characteristics. In low missing rate scenarios (η=0.1), the performance differences among all comparison models were not significant. However, as the missing rate η increased, baseline models showed significant performance degradation, reflecting the inherent limitations of traditional drug recommendation models in handling incomplete view data. In contrast, IMDR demonstrated stronger performance stability, with its evaluation metrics showing a notably gentler degradation curve as the missing rate increased. Particularly in high missing rate scenarios (η=0.9), IMDR maintained significant advantages across all key evaluation metrics, thoroughly confirming its superior robustness in addressing the challenge of missing views.

5.3 Performance Evaluation Against Incomplete Multi-view Learning Methods

This section aims to systematically evaluate the performance differences between the IMDR model and existing incomplete multi-view learning methods. The control group includes traditional statistical

[2]https://github.com/sunlabuiuc/PyHealth

Table 2: Performance of the IMDR model versus other IML models in MIMIC-III in terms of DDI rate, Jaccard and F1-score. The best and runner-up results are highlighted in bold and underline respectively.

Missing rate η	0.1			0.5			0.9			Avg		
Model	Jaccard ↑	F1-score ↑	DDI ↓	Jaccard ↑	F1-score ↑	DDI ↓	Jaccard ↑	F1-score ↑	DDI ↓	Jaccard ↑	F1-score ↑	DDI ↓
GAMENet+*avg*	0.4911	0.6397	0.0653	0.4414	0.5934	0.0695	0.3897	0.5403	0.0701	0.4419	0.5934	0.0696
GAMENet+*copy*	0.4899	0.6373	0.0611	0.4512	0.6031	0.0640	0.3837	0.5293	0.0653	0.4461	0.5962	0.0631
GAMENet+*random*	0.4876	0.6354	0.0650	0.4383	0.5893	0.0653	0.4097	0.5606	0.0610	0.4427	0.5935	0.0657
Transformer+*avg*	0.4938	0.6397	0.0643	0.4452	0.5952	0.0614	0.3966	0.5507	**0.0578**	0.4413	0.5904	**0.0597**
Transformer+*copy*	0.4913	0.6392	0.0668	0.4444	0.5936	0.0618	0.4194	0.5704	0.0620	0.4464	0.5949	0.0621
Transformer+*random*	0.4955	0.6437	**0.0607**	0.4302	0.5781	0.0629	0.3862	0.5325	0.0589	0.4368	0.5850	0.0609
DCP	0.4800	0.6260	0.0619	0.4207	0.5704	0.0673	0.3705	0.5234	0.0764	0.4224	0.5720	0.0638
IMVCDC	0.4919	0.6372	0.0612	0.4410	0.5892	0.0598	0.3960	0.5480	0.0628	0.4422	0.5907	0.0613
VIST	0.4983	0.6480	0.0632	0.4692	0.6198	0.0634	0.4266	0.5785	0.0654	0.4645	0.6155	0.0643
IMDR	**0.5030**	**0.6497**	0.0638	**0.4809**	**0.6270**	**0.0566**	**0.4728**	**0.6218**	0.0580	**0.4832**	**0.6303**	**0.0597**

Table 3: Ablation study on MIMIC-III (averaged across 9 missing rates η from 0.1 to 0.9) in terms of DDI rate, Jaccard, F1-score, PRAUC, ROCAUC and average number of medications.

Model	Jaccard↑	PRAUC↑	F1-score↑	ROCAUC↑	DDI↓	Avg.# of medications
$\mathcal{M}_0$: only Transformer	0.4525	0.7299	0.6024	0.9309	0.0615	24.1105
$\mathcal{M}_1$: + Code-level Drug Knowledge Infusion	0.4598	0.7326	0.6099	0.9323	0.0610	24.7123
$\mathcal{M}_2$: + Visit-level Multi-View Contrastive Learning	0.4652	0.7414	0.6159	0.9356	0.0624	24.0080
$\mathcal{M}_3$: + Bi-View Sequential Representation Learning	0.4733	0.7490	0.6239	**0.9378**	0.0612	22.0199
$\mathcal{M}_4$: + Gaussian-guided Curriculum Learning	**0.4832**	**0.7491**	**0.6303**	0.9359	**0.0597**	23.8911

imputation methods and recently proposed deep learning architectures. Experimental results in table 2 and table 5 demonstrate that IMDR achieves optimal performance across all mainstream evaluation metrics while maintaining a low DDI rate, outperforming baseline models with improvements of 4.03% in Jaccard similarity and 2.41% in F1-score. Among all baseline methods, only VIST achieved sub-optimal performance through its innovative high-order feature fusion mechanism and enhanced embedding representation, yet its overall effectiveness remains significantly below that of IMDR.

Notably, traditional imputation methods and classical generative models perform poorly when processing EHR data, particularly with the DCP model showing performance significantly below expectations. This phenomenon can be attributed to the inherent characteristics of EHR data: long-term temporal dependencies, complex interactions between medical entities, and heterogeneous features in clinical scenarios. In contrast, IMDR effectively addresses these challenges through its innovative dual-level architectural design: the code-level knowledge infusion mechanism effectively captures complex relationships between medical entities, the view-level cross-view alignment strategy ensures consistent representation of heterogeneous information, while the Gaussian-guided curriculum learning framework provides an effective paradigm for processing long-sequence temporal data. These results demonstrate that IMDR exhibits superior adaptability in handling missing view scenarios specifically in drug recommendation tasks compared to other IML models.

5.4 Ablation Studies

In our paper, we propose several key designs to handle missing views during drug recommendation. In this section, we will investigate how each component contribute to the final performance improvement. To systematically evaluate the contribution of each core component in the IMDR model, we designed a series of progressive ablation experiments on the MIMIC-III dataset. Through stepwise integration of functional modules, we observed the changes brought by each module from table 3.

From table 3, we can see that 1) Addition of the medical code-level drug knowledge infusion module(Section 4.1) not only improves the model's prediction accuracy but also effectively reduced the DDI rate; 2) Addition of multi-view contrastive learning(Section 4.2.2) further improves model performance; 3) Addition of bi-view sequential representation learning enhance the patient' representation learning. 4) Gaussian-guided curriculum learning brings significant performance gains. Overall, each designed module made a key contribution to the overall performance improvement.

5.5 Dynamic Sample Weighting Analysis

To intuitively demonstrate the regulatory effect of our Gaussian distribution-based curriculum learning strategy during training, Figure 3 illustrates the weight evolution curves of a typical difficult sample (subject id 107281) and an easy sample (subject id 120623). Specifically, subject id 120623 exhibits characteristics of an "easy sample" with low scores across multiple dimensions including view completeness, drug combination complexity, and clinical semantic complexity. During the early training phase, this sample is assigned a higher learning weight, which helps the model quickly establish basic understanding of the drug recommendation task. As training progresses and the model matures, its dependence on simple samples decreases, reflected in the gradual reduction of their weights. Conversely, subject id 107281 exemplifies a challenging sample characterized by incomplete view information, intricate drug interaction patterns, and elevated clinical semantic complexity. For such difficult samples, we employ a progressive learning strategy: assigning lower weights during early training, and then gradually

Subject id 107281 (hard sample)
Diagnoses: 41071、4280、43411、5750、43331、5070、2875、48241、5845、40391、0389、99592、51884、2749、4352、V5331、25040 **Procedures**: ~~311、5101、3723、8856、8844、0013、966、9604、9671、9672、9904、0017、9907~~ **Drugs**: V06D、R03B、B01A、B05X、R03A、C03C、C07A、N01A、C10A、R05C、A02B、N02B、A12B、A06A、N05C、J01X、J01M、C01C、P01A、A03F、A12C、V04C、C03B、C01D、J01C、N02A、B02B、A07X、C08C、A04A

Subject id 120623 (easy sample)
Diagnoses: V3001、V290、V502、V053 **Procedures**: 640、9955 **Drugs**: J01G、J01C

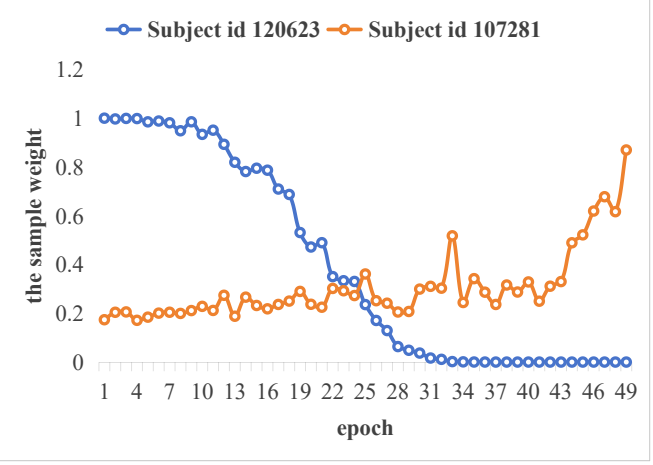


Figure 3: Weight Evolution for Easy and Hard Samples during Training.

increasing their importance after the model has established stable feature representation capabilities. This dynamic adjustment mechanism ensures that the model can progressively master clinical knowledge from simple to complex, thereby enhancing overall performance.

5.6 Clinical Practice

To validate the real-world applicability of IMDR, we conducted a comprehensive clinical practical analysis at Qilu Hospital of Shandong University involving five experienced physicians. The evaluation was designed as a double-blind assessment where the physicians were unaware of the source of the drug recommendations to ensure unbiased evaluation. Two types of experimental evaluations were performed to assess IMDR's clinical utility. In the first evaluation, physicians were asked to rate IMDR's drug recommendations on a 1-5 scale based on clinical appropriateness and treatment relevance, where IMDR achieved an average score of 4.4 out of 5, demonstrating strong clinical acceptance and indicating that the recommendations were considered highly reasonable by practicing clinicians. In the second comparative evaluation, physicians were presented with drug recommendations from IMDR alongside those from competing baseline methods in a blind comparison format, and remarkably, 4 out of 5 physicians consistently preferred IMDR's recommendations over alternative approaches. These compelling results from real-world clinical practice provide strong evidence of IMDR's practical value and deployment potential, confirming that our system generates clinically relevant and trustworthy drug recommendations that align well with physician judgment and clinical decision-making processes.

6 Conclusion

Our work focuses on addressing the challenge of missing views in drug recommendation tasks within real clinical environments by proposing a novel Incomplete Multi-View Drug Recommendation (IMDR) model. The model employs an innovative dual-level representation learning architecture that achieves medical code-level knowledge infusion and visit-level cross-view information alignment, combined with a Gaussian-guided adaptive curriculum learning strategy to enable progressive training from simple to complex samples. In evaluating missing view scenarios using two large-scale real-world medical datasets, MIMIC-III and MIMIC-IV, IMDR demonstrates significant performance improvements over existing state-of-the-art drug recommendation models, highlighting its exceptional adaptability in scenarios with missing views. Further comparative experiments validate IMDR's distinct advantages over advanced incomplete multi-view learning models, providing an effective solution for drug recommendation tasks in incomplete multi-view scenarios.

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A Details of Prompts

To comprehensively assess the semantic complexity of medical entities, we designed a set of specialized prompt templates and utilized advanced language model APIs to perform multi-dimensional scoring for different types of medical codes. Specifically, we developed dedicated evaluation templates for three categories of medical entities: medications, diagnoses, and procedures.

Input Prompt Template for ATC3

```
    Rate the complexity of the following medication
on a scale of 1-10, considering:
        - Administration complexity
        - Side effect profile
        - Monitoring requirements
        - Drug interactions potential
        - Treatment duration

    Medication: 'STOMATOLOGICAL PREPARATIONS'

    Return only a number between 1-10, where:
        1: Simple, safe, minimal monitoring
        10: Complex, high-risk, intensive monitoring
required
```

Figure 4: Sample Complexity Rating Template: STOMATOLOGICAL PREPARATIONS (ATC3 Code: 0A01).

B Experimental Settings

B.1 Dataset

We utilized two datasets, MIMIC-III and MIMIC-IV, and performed unified data processing using the open-source benchmark framework PyHealth. The statistical characteristics of the processed datasets are summarized in the table 4. These datasets are large-scale de-identified healthcare records for ICU patients, publicly available at https://mimic.physionet.org/.

Table 4: Statistics of the Datasets

Items	MIMIC-III	MIMIC-IV
# of patients	5449	32633
# of visits	14141	105441
# of diagnosis	4491	17267
# of procedure	1412	8923
# of medication	193	199
avg # of visits	2.5952	3.2311
avg # of diagnosis	31.6520	47.1405
avg # of procedure	9.4601	8.0635
avg # of medication	29.7427	24.6683

During the data preprocessing phase, we standardized the diagnoses and procedures coding according to the International Classification of Diseases, Ninth Revision (ICD-9), and normalized the medication data using the third level of the Anatomical Therapeutic Chemical (ATC) classification system. Additionally, we referenced the DDI knowledge base from [16] and adopted their methodology to construct the DDI matrix for evaluating the safety of drug combinations.

B.2 Baselines

To comprehensively evaluate the performance of the IMDR model, we designed two categories of comparative experiments. The first category compares against mainstream drug recommendation baseline models:

- **MLP** is a simple neural network architecture. It does not perform sequence modeling on $H_t^{(v_i)}$, meaning it does not capture temporal relationships between visits. MLP directly feeds the obtained patient representations into a feed-forward neural network to generate the final output.
- **Transformer** [16] is a neural network based on self-attention mechanisms. Unlike MLP, Transformer employs self-attention mechanisms to model $H_t^{(v_i)}$ and $H_t^{(v_j)}$ separately, enabling it to capture temporal relationships between different patient visits. Similarly, Transformer feeds the obtained patient representations into a feedforward neural network to generate the final output.
- **RETAIN** [3] is an interpretable longitudinal prediction model based on RNN and hierarchical attention mechanisms. RETAIN utilizes RNN and attention mechanisms to predict future medication combinations based on patients' historical visit records.
- **GAMENet** [16] is a longitudinal prediction model for drug combination recommendation. GAMENet innovatively combines RNN, memory networks, and graph neural networks, leveraging knowledge from drug co-occurrence graphs and DDI graphs to better capture drug interactions and improve prediction accuracy.
- **SafeDrug** [25] is a model specifically designed for predicting drug combination safety. SafeDrug integrates drug molecular structure information and drug interaction information, utilizing graph neural networks to learn relationships between drugs, thereby more accurately identifying potentially unsafe drug combinations.
- **MICRON** [24] is a two-stage drug recommendation model. MICRON employs a recurrent residual learning model to capture medication changes between consecutive visits and makes recommendations based on these changes and patients' recent medication combinations.
- **MoleRec** [26] significantly improves drug combination recommendation by utilizing molecular substructure interactions and patient substructure correlations to identify efficacy-driving substructures.

All these models were implemented on the open-source framework PyHealth, ensuring experimental fairness and reproducibility through unified data preprocessing methods. However, since these traditional drug recommendation models were not designed to handle missing views, we introduced a second category of baselines to conduct a more comprehensive performance evaluation, comparing IMDR with advanced incomplete multi-view learning methods:

- Traditional statistical view completion methods: These include **zero-filling**, **mean-filling** [8], **View-copying**, and **random-filling** strategies, each employing their respective approaches to complete missing view representations.

Table 5: Performance comparison of IMDR and other IML models on MIMIC-IV where $\eta = 0.5$. Metrics include Jaccard, PR AUC, F1, ROC AUC, DDI, and average number of medications.

Model	Jaccard ↑	PRAUC ↑	F1-score ↑	ROCAUC ↑	DDI ↓	Avg.# of medications
GAMENet+*random*	0.4453	0.7132	0.5916	0.9348	0.0664	18.41
Transformer+*random*	0.4179	0.7010	0.5569	0.9295	0.0619	14.05
DCP	0.3378	0.6124	0.4735	0.8946	**0.0404**	20.54
IMVCDC	0.4385	0.6939	0.5792	0.9230	0.0591	18.97
VIST	0.4666	0.7338	0.6089	0.9395	0.0625	18.01
IMDR (Ours)	**0.4750**	**0.7444**	**0.6133**	**0.9413**	0.0556	17.07

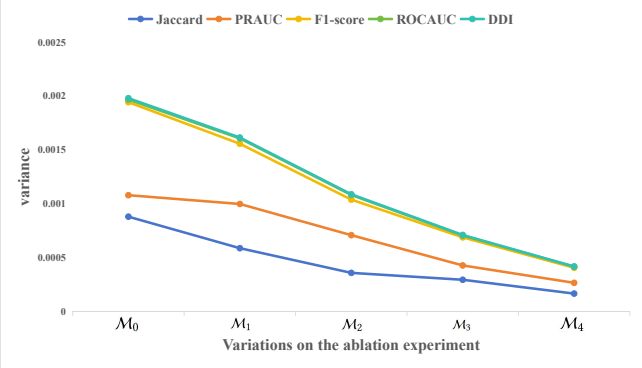


Figure 5: Model Performance Variance after Component Addition. The sequential addition of model components demonstrates IMDR's robust performance across different missing rates (η from 0.1 to 0.9), as evidenced by the consistent decrease in performance variance across all metrics. Most notably, the F1-score and DDI rate variances show substantial improvement in stability, decreasing from approximately 2e-3 to below 5e-4, while both the Jaccard similarity and PRAUC maintain steady performance with reduced variance. This systematic reduction in performance variance validates our architectural design and confirms that IMDR achieves robust training and reliable drug recommendations even in challenging scenarios with significant missing view data.

- **DCP**[10] is an incomplete multi-view representation learning model that employs dual contrastive prediction, utilizing contrastive learning to capture consistency features between different views while recovering missing views through bidirectional prediction.
- **IMVCDC** [4] is a three-stage incomplete multi-view clustering model. This method first extracts latent view representations using autoencoders, then completes view information in the latent space through diffusion models, and finally achieves multi-view clustering based on spectral contrastive loss.
- **VIST** [14] proposes a dual-layer view-category interactive shared Transformer architecture, achieving efficient incomplete multi-view multi-label classification through an integrated missing view generation mechanism combining KNN and multivariate Gaussian distribution, along with a view-category consistency-based contrastive learning strategy.

To objectively evaluate model performance, we employed six evaluation metrics: Jaccard similarity (Jaccard) to measure the overlap between predicted results and ground truth labels; Precision-Recall Area Under Curve (PR-AUC) and Receiver Operating Characteristic Area Under Curve (AUC-ROC) to assess the model's overall discriminative capability; Average F1 score to comprehensively evaluate precision and recall; Drug-Drug Interaction rate (DDI rate) to quantify the safety of recommendations; and Average number of medications.

B.3 Experimental Details

To ensure experimental reliability and practicality, we constructed a realistic clinical scenario based on the MIMIC series datasets. The datasets were partitioned using stratified random sampling: 70% for model training, 15% for parameter validation, and the remaining 15% for performance testing. To systematically evaluate model performance under varying degrees of view missing, we constructed three types of samples in each data subset with a ratio of 1-η:η/2:η/2: complete view samples, samples with missing diagnosis view, and samples with missing procedure view. By uniformly selecting 9 missing rates η from the interval [0.1, 0.9], we were able to comprehensively assess the model's robustness under different degrees of missing data.

For model training details, IMDR adopts a unified end-to-end training paradigm capable of simultaneously processing both complete view samples and missing view samples. For complete view samples, the model follows the training process described in the methodology section; for missing view samples, we replace the missing view with the observable view (i.e., $V_j = V_i$) during the cross-attention step, while maintaining the same processing approach as the methodology for other steps. This data augmentation technique enhances the model's adaptability to different degrees of view missing, thereby ensuring the continuity and stability of model training.

On the MIMIC-III dataset, IMDR's main parameters were set as follows: batch size = 256, $\alpha = \beta = 0.1$, $\tau = 0.5$, $\rho = 1.0$, $\delta = 0.1$. The network architecture employed single-layer self-attention and cross-attention layers with 8 attention heads, along with two layers of graph convolutional networks for feature extraction. A dropout rate of 0.1 was set in the Transformer block, and both embedding and hidden dimensions were uniformly set to 256. During training, we used the Adam optimizer with a learning rate of $5e-3$, selected the optimal model configuration based on validation set performance over 50 training epochs, and conducted final performance evaluation on the test set.